ENGINEERING CHEMISTRY

R23 Regulation • JNTUK Kakinada • Common for All Branches

Year / Sem	Theory Credits	Lab Credits	Total Credits
1st Year – 2nd Semester	3	1	4

DETAILED SYLLABUS

UNIT I — STRUCTURE AND BONDING

Topic: *Quantum Chemistry & Molecular Orbital Theory*

Focus Areas:
- Basics of Quantum Chemistry
- Schrodinger wave equation
- Molecular Orbital Theory (MOT)
- Bond order calculations
- Energy diagrams (O2, CO, etc.)
- Pi-bonding in benzene and butadiene

UNIT II — MODERN ENGINEERING MATERIALS

Topic: *Semiconductors, Superconductors & Nanomaterials*

Focus Areas:
- Semiconductors (concept and applications)
- Superconductors
- Supercapacitors
- Nanomaterials (CNT, graphene, fullerenes)
- Applications in engineering

UNIT III — ELECTROCHEMISTRY AND APPLICATIONS

Topic: *Electrochemical Cells, Titrations & Sensors*

Focus Areas:
- Electrochemical cells
- Nernst equation (numericals)
- Conductivity and conductometric titrations
- Potentiometric titrations
- Electrochemical sensors

UNIT IV	FUELS AND ENGINEERING MATERIALS
	Topic: *Fuels, Petroleum, Lubricants, Cement & Composites*
	Focus Areas: • Fuels and calorific value • Coal analysis (proximate and ultimate) • Petroleum refining • Octane and Cetane numbers • Lubricants (properties and applications) • Cement (setting and hardening) • Composites and refractories

UNIT V	SURFACE CHEMISTRY AND NANOMATERIALS
	Topic: *Colloids, Adsorption Isotherms & Nanomaterial Synthesis*
	Focus Areas: • Colloids and micelles • Adsorption isotherms (Freundlich, Langmuir) • Nanomaterials synthesis • Applications in Medicine, Catalysis, and Sensors

ENGINEERING CHEMISTRY LAB (Practical)

■	**Water Hardness Determination** Estimation of temporary and permanent hardness
■	**Dissolved Oxygen** Winkler method for DO measurement
■	**Polymer Preparation** Synthesis of Bakelite in laboratory
■	**Conductivity Experiments** Measurement and conductometric titrations
■	**Cement and Iron Estimation** Quantitative analysis techniques

COURSE OBJECTIVES

01	Understand fundamental chemical concepts applied in engineering contexts

02	Learn about materials, fuels, and electrochemistry with real-world relevance
03	Apply chemistry principles to solve real-world engineering problems

■ IMPORTANT EXAM TIPS

• Focus on **Numericals** — Nernst equation, calorific value calculations
• Focus on **Applications-based questions** for all units
• This subject falls under **Basic Science and Humanities (BS&H)** category
• Common for ALL branches: CSE, ECE, EEE, Civil, Mechanical, etc.